CSE 499 Project Plan

Project Name: Project 7 – To do List web and page videos of exercises

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Requirements:

List your project requirements with accompanying user stories. All the “core” requirements must be completed. Optional requirements are features you would like to add if time permits. For each requirement list the user stories you will be addressing.

Required

Use this table to document the features of your project. **You must have at least 4 requirements.** Modify the table as needed.

<table>
<tr><th>Requirement</th><th>C/E*</th><th>Description</th></tr>
<tr><td rowspan="2">1. Task Management (CRUD Operations)</td><td rowspan="2">Core</td><td>A web interface built with HTML, CSS, and JavaScript will allow users to interact with the system dynamically (rendering task cards, handling events).</td></tr>
<tr><td>User Stories
<table>
<tr><th>Name</th><th>Description</th></tr>
<tr><td>Create Task</td><td>As a user, I want to add a new task so that I can track my work.</td></tr>
<tr><td>View Tasks</td><td>As a user, I want to see all my tasks so that I can manage them.</td></tr>
<tr><td>Delete Task</td><td>As a user, I want to remove tasks so that I can clear completed work.</td></tr>
</table></td></tr>
<tr><td rowspan="2">2. Relational Database Integration</td><td rowspan="2">Core</td><td>The application will use a relational schema (Tasks and Users tables) connected via foreign keys to persist data using SQL (SQLite).</td></tr>
<tr><td>User Stories
<table>
<tr><th>Name</th><th>Description</th></tr>
<tr><td>Save Data</td><td>As a user, I want my tasks saved so they persist after closing the app.</td></tr>
<tr><td>Retrieve Data</td><td>As a user, I want my tasks loaded automatically when reopening.</td></tr>
<tr><td>Structured Data</td><td>As a developer, I want relationships between tables for better data organization.</td></tr>
</table></td></tr>
<tr><td rowspan="2">3. Web Interface (Frontend UI)</td><td rowspan="2">Core</td><td>A web interface built with HTML, CSS, and JavaScript will allow users to interact with the system dynamically (rendering task cards, handling events).</td></tr>
<tr><td>User Stories</td></tr>
</table>

<table>
<tr><td></td><td></td><td><table><tr><th>Name</th><th>Description</th></tr><tr><td>Display Tasks</td><td>As a user, I want to visually see my tasks in the browser.</td></tr><tr><td>Interactive UI</td><td>As a user, I want buttons to interact with tasks easily.</td></tr><tr><td>Dynamic Updates</td><td>As a user, I want actions reflected without reloading the page.</td></tr></table></td></tr>
<tr><td rowspan="2">4. Backend API (Web Services)</td><td rowspan="2">Core</td><td>A backend (Node.js or Java Spring Boot) will expose REST APIs (GET, POST, DELETE) to connect the frontend with the database.</td></tr>
<tr><td>User Stories<table><tr><th>Name</th><th>Description</th></tr><tr><td>API Create</td><td>As a developer, I want an endpoint to create tasks.</td></tr><tr><td>API Retrieve</td><td>As a developer, I want an endpoint to fetch tasks.</td></tr><tr><td>API Delete</td><td>As a developer, I want an endpoint to remove tasks.</td></tr></table></td></tr>
<tr><td rowspan="2">5. Advanced Features (Optional Enhancements)</td><td rowspan="2">Enhancement</td><td>Additional features such as task status updates, filtering, and sorting using SQL expressions and conditions.</td></tr>
<tr><td>User Stories<table><tr><th>Name</th><th>Description</th></tr><tr><td>Filter Tasks</td><td>As a user, I want to filter tasks by status or date.</td></tr><tr><td>Update Status</td><td>As a user, I want to mark tasks as completed.</td></tr><tr><td>Sort Tasks</td><td>As a user, I want tasks ordered by priority or due date.</td></tr></table></td></tr>
<tr><td rowspan="2">6. Provide a page where users can access and watch instructional videos.</td><td rowspan="2">Enhancement</td><td>A dedicated web page that displays exercise/tutorial videos, allowing users to watch and learn how to use the application and understand key programming concepts.</td></tr>
<tr><td>User Stories<table><tr><th>Name</th><th>Description</th></tr><tr><td>Watch Exercise Videos</td><td>As a user, I want to watch videos directly on the page so that I can learn how to use the application and understand the exercises.</td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table></td></tr>
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</table>

*C=Core, E=Enhancement

Project Schedule:

Create a rough project schedule broken down by sprint. For each sprint list the milestones you will meet. This is a rough guideline for future planning, you will make a more detailed plan at the first of each sprint. You will not be held accountable to this schedule as it will likely change as the class progresses.

Sprint	Milestone(s)
1	Set up environment, install SQLite, create initial database schema, basic SQL queries.
2	Implement CRUD operations and JavaScript logic to connect to database.
3	Build frontend interface and integrate backend (Node.js → Java Spring Boot migration).
4	Final testing, debugging, documentation, CI/CD setup, and video demo.

Project Architecture

Describe the architecture of your application (For example: web, mobile, client-server, n-tier, etc.).

Frontend (Client): HTML, CSS, JavaScript

Backend (Server): Java Spring Boot REST API

Database: SQLite relational database

Technology

Describe the technology you anticipate using (For example: programming languages, platforms, databases, etc.).

Languages: JavaScript, Java

Frontend: HTML, CSS, JavaScript

Backend: Node.js (initial) → Java Spring Boot

Database: SQLite/ Youtube videos

ORM: Spring Data JPA

Tools: GitHub, Maven, REST APIs